Flowintel

Installation test and analysis of IoC start and end dates per threat-intelligence source

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Software: Flowintel 4.0.0 (commit 8a8e8c2, 2026-10-28)

Test platform: Proxmox VE 9.2 – VM 106 'flowintel-test', Ubuntu 24.04.5 LTS

Sources evaluated: 108 threat-intelligence feeds (MISP default feed catalogue)

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Prepared with: Claude Code

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1. Executive summary

Flowintel 4.0.0 was installed from the official repository on a new Ubuntu 24.04 virtual machine on the Proxmox host, using the project's install.sh with PostgreSQL 16 and Valkey 7.2. The web application, the REST API, the MISP modules and the bundled test data all worked. The automated test suite (654 tests) passed 653; the single failure is a concurrency test showing that two simultaneous duplicate alert ingests lose one occurrence count.

Flowintel does not collect threat intelligence itself; IoCs enter it as MISP events. All 108 sources in the MISP project's default feed catalogue were therefore downloaded and imported through the Flowintel API (POST /api/case/create_with_event). 79 sources could be used (14 in MISP format, 65 CSV/free-text); 29 were unavailable or returned no data. In total 3,521,210 IoCs were stored in 17,045 cases.

For every source the start date (first_seen) and end date (last_seen) of each IoC were counted, statistically described and compared with the raw feed. Only MISP-format sources carry IoC dates: of 1,951,282 IoCs, 291,692 (14.9%) have a start date, 315,493 (16.2%) an end date and 21,239 (1.1%) both. 4 of the 14 MISP sources provide no IoC dates at all, and none of the 1,569,928 CSV/free-text IoCs has a date in Flowintel, even where the feed lines contain dates.

Flowintel preserves attribute dates exactly (0 differences between source and Flowintel after the workaround described below). Two defects affect the result, however: (1) IoC values longer than 255 characters make the import fail with HTTP 500 and leave incomplete cases, which in the default configuration lost 338,907 IoCs (9.6%) from 15 sources; (2) first_seen/last_seen set at MISP-object level are not stored by Flowintel.

Key figures

Measure	Value
Sources in catalogue / usable	108 / 79
IoCs imported into Flowintel	3,521,210
Cases created	17,045
IoCs from MISP-format sources	1,951,282
… with first_seen (start date)	291,692 (14.9%)
… with last_seen (end date)	315,493 (16.2%)
… with both dates	21,239 (1.1%)
… with no date	1,365,336 (70.0%)
IoCs from CSV/free-text sources (no dates in Flowintel)	1,569,928
Source vs Flowintel mismatches (after workaround)	0
IoCs lost with default schema (value > 255 chars)	338,907 (9.6%)
Automated test suite	653 of 654 passed (1 failed)

Main findings

- Only 2 sources (NOCACTI: Intrusion Feed, NOCACTI: Adversary Infrastructure Feed) give both dates for every IoC; Infoblox gives a start date for all 269,362 IoCs but never an end date.
- CIRCL OSINT, the largest source (588,220 IoCs), has an end date on 51% of IoCs, almost all from the daily 'Maltrail IOC' events where last_seen is the publication day, and both dates on only 1.1%.
- Rösti, PrecisionSec, TweetFeed and SiberKapan (631,295 IoCs) carry no IoC dates; PrecisionSec puts dates in first-seen/last-seen object attributes instead (14,835 / 4,792).

- CSV/free-text feeds never yield dates in Flowintel (as in MISP), although 21 of them have dates on most lines (e.g. dataplane.org, abuse.ch SSLBL, Phishtank).
- Flowintel bug: misp_attribute.value is VARCHAR(255); longer values abort the request with HTTP 500 and the case is left half-imported because each attribute is committed separately.
- Flowintel does not store object-level first_seen/last_seen (e.g. all 1,801 dated objects of NOCACTI Intrusion, 859 of Rectifyq 2026).

2. Scope and approach

The task had three parts: (1) install a new Ubuntu VM on Proxmox; (2) install and test Flowintel on it; (3) import all threat-intelligence sources containing IoCs and evaluate, for each source, the information available and how many IoCs have a start date and an end date, with statistics.

Source catalogue. Flowintel has no feed catalogue of its own; its threat-intelligence model is MISP. The authoritative public list of MISP feeds, feed-metadata/defaults.json from the MISP project (108 feeds), was used as the definition of 'all sources'.

Import path. MISP-format feeds were downloaded in full (manifest.json plus every event) and each event was sent unchanged to POST /api/case/create_with_event, one case per event. CSV and free-text feeds were parsed with the value columns and exclusion rules defined for each feed in the catalogue, the same way MISP does it, and sent as one event per feed (split into cases of at most 50,000 IoCs). MISP does not map dates from CSV/free-text feeds either, so these IoCs carry no first_seen/last_seen.

Measurement. Statistics were computed twice: directly from the downloaded feed files (the 'source' view) and from the Flowintel PostgreSQL database through a read-only query (the 'Flowintel' view). The two views were compared source by source.

2.1 Definitions

- **IoC –** A MISP attribute stored by Flowintel (table misp__attribute), standalone or inside a MISP object.
- **Start date –** Attribute first_seen.
- **End date –** Attribute last_seen.
- **Both / none –** IoC with first_seen and last_seen / with neither.
- **Lifetime –** last_seen − first_seen in days, for IoCs that have both; median, mean, 90th percentile and maximum are reported.
- **Inverted –** IoC whose last_seen is earlier than its first_seen (none were found).
- **Zero-length –** IoC whose first_seen equals last_seen.

3. Test environment

Component	Value
Hypervisor	Proxmox VE 9.2.24 (kernel 7.0.14-26-pve), MacBook Pro 15,1
Virtual machine	VM 106 'flowintel-test': 4 vCPU (host CPU), 8 GB RAM, 60 GB disk (LVM-thin), virtio NIC on vmbr0 (NAT, DHCP 10.10.98.131)
Guest OS	Ubuntu 24.04.5 LTS server cloud image, kernel 6.8.0-147-generic, cloud-init user 'flowintel'
Flowintel	4.0.0, commit 8a8e8c2 (2026-10-28), development launch (./launch.sh -l) on port 7006
Database	PostgreSQL 16.15 (database 'flowintel', 1.6 GB after import)
Session store	Valkey 7.2.13 with password
Runtime	Python 3.12.3 venv (uv), Flask 3.1.3, SQLAlchemy 2.1.6, PyMISP 2.5.34.4, misp-modules 3.1.1, Node.js 26.13.0, Bun, pandoc 3.7
Reference data	181 MISP taxonomies (13,051 tags), 131 galaxies (54,731 clusters), MISP objects submodule
Source catalogue	MISP feed-metadata/defaults.json (branch 2.5), 108 feeds, downloaded 29 Oct 2026 (1.6 GB)

4. Installation of Flowintel

Installation followed the README quick start. The only choices made were PostgreSQL as database (README prerequisite) and binding the development server to all interfaces so the UI is reachable from the host network. Configuration is supplied as environment variables (the application reads no .env file).

4.1 Steps performed

- Created VM 106 from the Ubuntu 24.04 cloud image with cloud-init (SSH key, DHCP) and updated the system.

  qm create 106 … ; qm set 106 --scsi0 local-thin-storage:0,import-from=noble-server-cloudimg-amd64.img ; apt full-upgrade
- Installed PostgreSQL 16 and Valkey; created the database and role; set a Valkey password.

  apt install postgresql valkey-server ; CREATE USER flowintel … ; CREATE DATABASE flowintel OWNER flowintel
- Cloned the repository and copied the default configuration files.

  git clone https://github.com/flowintel/flowintel ; cp conf/config.py.default conf/config.py ; cp conf/config_module.py.default conf/config_module.py
- Exported the configuration as environment variables (SECRET_KEY, DB_*, VALKEY_*, FLOWINTEL_APP_HOST=0.0.0.0).

  ~/flowintel.env (mode 600)
- Ran the official installer: system packages, pandoc, Node/nvm, Mermaid CLI, Python venv (uv), Bun assets, git submodules, database initialisation, taxonomies, galaxies, MISP modules and default cases.

  ./install.sh (exit code 0, about 40 minutes)
- Started the application.

  ./launch.sh -l (Flask, misp-modules, notification and MISP-sync workers in screen sessions)
- Loaded the bundled community test data through the API.

  ./launch.sh -tdc <admin_api_key>
- Installed the development requirements and ran the automated test suite.

  uv pip install -r requirements-dev.txt ; FLOWINTEL_APP_ENV=testing python -m pytest -n 4 -p no:randomly --cov=./app

4.2 Issues and deviations found

#	Observation	Impact / action taken
1	misp__attribute.value is VARCHAR(255). Any IoC value longer than 255 characters makes POST /api/case/create_with_event fail with HTTP 500 (StringDataRightTruncation on PostgreSQL). Because every attribute is committed separately, the case remains in the database half-imported. 3,530 cases (MISP events and CSV/free-text chunks) were affected; 338,907 IoCs (9.6%) were missing.	Critical for TI imports. Test workaround: column changed to TEXT, the B-tree index on value replaced by a hash index, incomplete cases deleted and re-created through the API (3,530 cases). Afterwards 0 differences against the sources. SQLite does not enforce the length, so the problem only appears with PostgreSQL/MariaDB.
2	MISP object-level first_seen/last_seen are not stored (table case__misp__object has no date columns).	Date information lost for objects, e.g. all 1,801 dated objects of NOCACTI Intrusion and 859 of Rectifyq 2026. Not worked around.
3	Bulk import is slow: one INSERT + COMMIT + SELECT of the case per attribute, about 90 IoCs/s per application process (database initialisation of taxonomies and galaxies took about 35 minutes for the same reason).	Test workaround: PostgreSQL synchronous_commit=off and four extra application processes on ports 7007–7010 sharing the database (about 290 IoCs/s). The full import of 3.5 million IoCs took about 3 hours.
4	No index on misp__attribute.case_misp_object_id.	Deleting cases/objects scans the whole attribute table. Index added for the test.
5	install.sh installs only requirements.txt, so ./launch.sh -t fails with 'No module named pytest'.	Installed requirements-dev.txt.

#	Observation	Impact / action taken
6	launch.sh uses the shebang '#!/bin/bash -i' (interactive shell) and therefore reads ~/.bashrc. With DB_* variables exported from .bashrc the test suite connects to PostgreSQL ('database flowintel_test_master does not exist') and every test errors at setup.	Configuration removed from .bashrc for the test run. The '_test_' name guard in tests/conftest.py prevents the production database from being dropped, but the behaviour is fragile.
7	Test suite: test_concurrent_duplicate_ingests_count_every_occurrence fails (occurrence_count 2, expected 3).	Concurrent ingests of the same alert can lose an occurrence count (lost update). Minor for this analysis; worth reporting upstream.
8	install.sh runs 'nvm install node 20.19.2', which installs the latest Node.js (26.13.0), not 20.19.2.	No functional impact seen; Node version not pinned as intended.
9	The development launch runs Flask in DEBUG mode with the Werkzeug debugger; with FLOWINTEL_APP_HOST=0.0.0.0 it is reachable from the network.	Acceptable only on an isolated test network (here NAT behind the Proxmox host). Use ./launch.sh -p with gunicorn behind nginx for real use.
10	Application processes grew to about 1.4 GB RAM each during the bulk import.	Observation; the 8 GB VM was sufficient.
11	'apt install valkey' in install.sh resolves to valkey-server on Ubuntu 24.04.	No issue.

4.3 Functional verification

Check	Result
Web UI login page (from the Proxmox host)	HTTP 200 at http://10.10.98.131:7006
Default administrator (admin@admin.admin)	Created by the installer; API key used for all API calls
Taxonomies, galaxies, MISP objects, MISP modules	Loaded (181 taxonomies, 131 galaxies); misp-modules server started by launch.sh
Bundled test data (-tdc)	Organisations and users created; the 4 template cases were already created by install.sh (a second -tdcc would only duplicate titles)
API case creation from MISP events	Works (16,955 events imported); fails with HTTP 500 for values > 255 characters (see 4.2)
Preservation of attribute first_seen/last_seen	Exact for all 14 MISP-format sources (0 differences)
Automated test suite (./launch.sh -t, 654 tests)	653 passed, 1 failed (tests/alerts/test_alert_occurrence_count_race.py: occurrence_count 2 instead of 3 after concurrent duplicate ingests, a lost-update race). Run with 4 parallel workers (pytest -n 4, SQLite test database, 1 h 50 min); line coverage 44%.

5. Threat-intelligence sources

The catalogue contains 108 feeds: 18 MISP-format, 47 CSV and 43 free-text. Downloads were made from the VM on 29 October 2026 with up to two retries; TweetFeed needed a slow sequential retry because of rate limiting.

Feed format	Listed	Downloaded	Imported into Flowintel	IoCs imported
freetext	43	38	34	1,099,602
misp	18	14	14	1,951,282
csv	47	31	31	470,326
Total	108	83	79	3,521,210

Table 5.1 – Source catalogue: MISP project default feed list (feed-metadata/defaults.json), all 108 entries.

5.1 Sources that could not be used

The following sources could not be used. Most are discontinued or moved (abuse.ch returns 410 Gone for all legacy export URLs), require registration, or returned an HTML page or empty content.

ID	Source	Format	Reason
F000	ELLIO: IP Feed (Community version)	freetext	HTML page returned instead of feed data
F008	ip-block-list - snort.org	freetext	HTML page returned instead of feed data
F014	IPs from High-Confidence DGA-Based C&Cs Actively Resolving - requires	csv	HTTPError: HTTP Error 403: Forbidden
F015	Domains from High-Confidence DGA-based C&C Domains Actively Resolving	csv	HTTPError: HTTP Error 403: Forbidden
F031	All current domains belonging to known malicious DGAs	csv	HTTPError: HTTP Error 403: Forbidden
F037	This list contains all domains - A list for administrators to prevent	freetext	HTTPError: HTTP Error 403: Forbidden
F038	This list contains all optional domains - An additional list for admin	freetext	HTTPError: HTTP Error 403: Forbidden
F039	This list contains all browser mining domains - A list to prevent brow	freetext	HTTPError: HTTP Error 403: Forbidden
F040	URLHaus Malware URLs	csv	HTTPError: HTTP Error 410: Gone
F041	CyberCure - IP Feed	csv	HTTPError: HTTP Error 520: <none>
F042	CyberCure - Blocked URL Feed	csv	HTTPError: HTTP Error 520: <none>
F043	CyberCure - Hash Feed	csv	HTTPError: HTTP Error 520: <none>
F049	Panels Tracker	csv	HTTPError: HTTP Error 404: Not Found
F058	DigitalSide Threat-Intel OSINT Feed	misp	URLError: <urlopen error timed out>
F060	Malware Bazaar	csv	HTTPError: HTTP Error 410: Gone
F061	PhishScore	csv	HTTPError: HTTP Error 404: Not Found
F062	Threatfox	misp	HTTPError: HTTP Error 410: Gone
F063	MalwareBazaar	misp	HTTPError: HTTP Error 410: Gone
F064	URLhaus	misp	HTTPError: HTTP Error 410: Gone
F068	threatfox indicators of compromise	csv	HTTPError: HTTP Error 410: Gone
F071	List of malicious hashes	csv	URLError: <urlopen error [Errno 111] Connection refused>
F072	Shreshta: Newly Registered domain names(NRD) - 1 week (Community polic	freetext	empty response
F073	Shreshta: Newly Registered domain names (NRD) - 1 month (Community pol	freetext	empty response
F084	Phishing.Database - New domains of today	csv	HTTPError: HTTP Error 503: Service Unavailable
F085	Phishing.Database - New IPs of today	csv	HTTPError: HTTP Error 503: Service Unavailable

ID	Source	Format	Reason
F086	Phishing.Database - New URLs of today	csv	HTTPError: HTTP Error 503: Service Unavailable
F088	hideNseek LAB - Threat Intelligence JSON	freetext	HTTPError: HTTP Error 525: <none>
F089	hideNseek LAB - CSV Threat Feed	csv	HTTPError: HTTP Error 525: <none>
F090	hideNseek LAB - IP Blocklist	freetext	HTTPError: HTTP Error 525: <none>

6. Results – MISP-format sources (IoCs with native dates)

MISP-format sources are the only ones whose IoCs can carry start and end dates. Table 6.1 gives the counts per source as stored in Flowintel; they are identical to the counts computed from the raw feed files (section 6.4).

6.1 Start and end dates per source

ID	Source	Events	IoCs	first_seen	last_seen	Both	None	% with any date
F001	CIRCL OSINT Feed	1,681	588,220	6,894	300,638	6,384	287,072	51.2%
F002	The Botvrij.eu Data	435	28,915	329	5	5	28,586	1.1%
F074	Infoblox-Threat-Intelligence	47	269,362	269,362	0	0	0	100.0%
F083	Rösti	10,491	428,536	0	0	0	428,536	0.0%
F091	NOCACTI: Intrusion Feed	3	6,419	6,419	6,419	6,419	0	100.0%
F092	NOCACTI: Adversary Infrastructure Feed	1	1,060	1,060	1,060	1,060	0	100.0%
F093	Rectifyq 2024	551	79,955	255	0	0	79,700	0.3%
F094	Rectifyq 2025	1,283	224,292	2	1	1	224,290	0.0%
F095	Rectifyq 2026	867	70,867	7,118	7,118	7,118	63,749	10.0%
F096	Rectifyq ICS-OT	101	7,274	17	17	17	7,257	0.2%
F097	Rectifyq Malaysia	214	43,623	236	235	235	43,387	0.5%
F098	PrecisionSec OSINT	856	90,353	0	0	0	90,353	0.0%
F099	SiberKapan	60	14,427	0	0	0	14,427	0.0%
F102	TweetFeed	365	97,979	0	0	0	97,979	0.0%

Table 6.1 – IoCs (MISP attributes, standalone and inside objects) stored by Flowintel per source, and how many carry first_seen / last_seen.

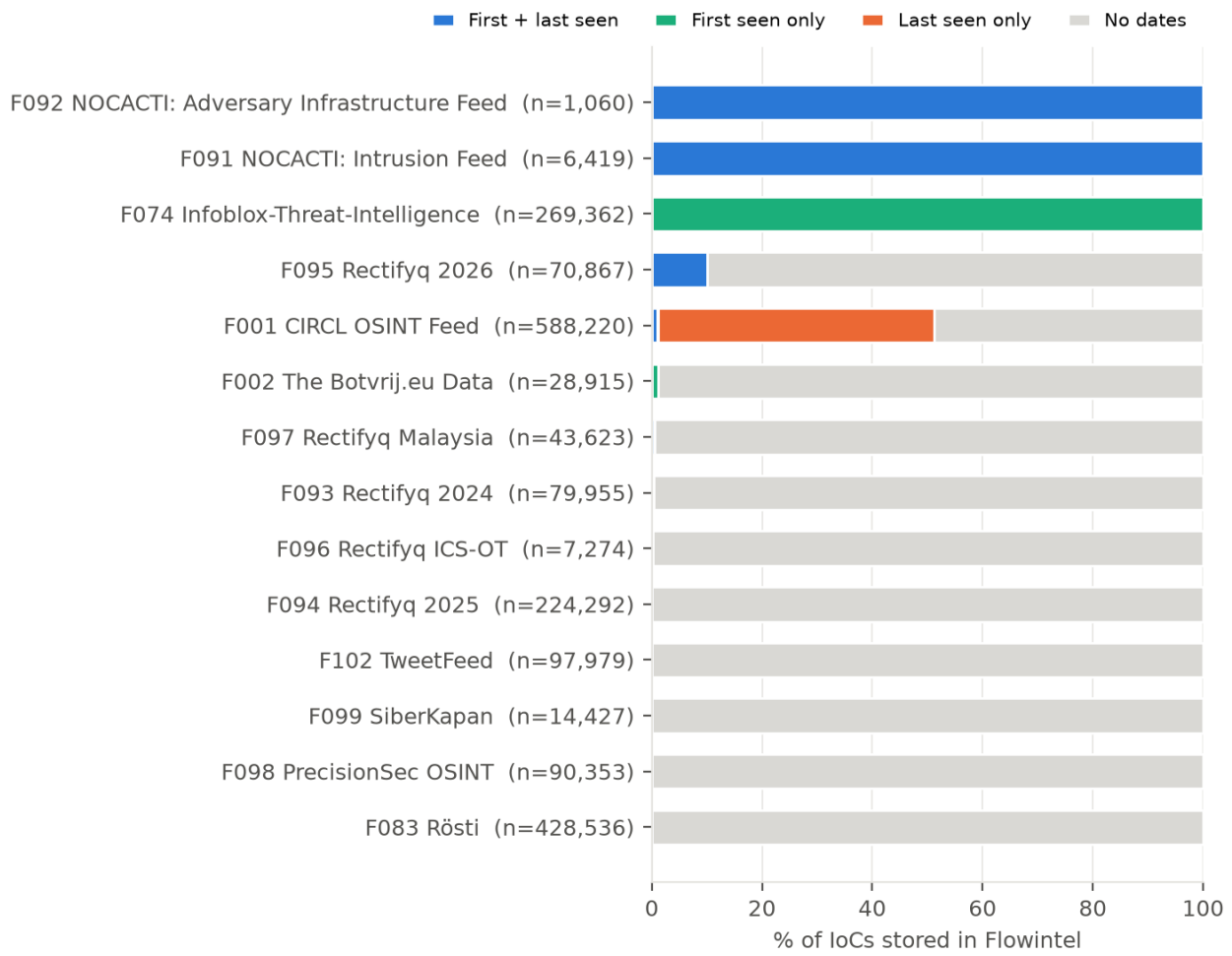


Figure 6.1 – Share of IoCs per source with start date, end date, both, or neither (as stored in Flowintel).

6.2 Date ranges and IoC lifetime

For IoCs that have both dates, the interval last_seen − first_seen was computed. No inverted ranges were found in any source. Several sources mainly give zero- or near-zero-length intervals (first and last observation within minutes), so the dates describe a single sighting rather than an active period.

ID	Source	first_seen range	last_seen range	Median (d)	P90 (d)	Max (d)	Zero-length	Inverted
F001	CIRCL OSINT Feed	2014-05-23 → 2026-05-13	2020-01-27 → 2026-08-14	44	56	56	574	0
F002	The Botvrij.eu Data	2014-05-23 → 2024-06-01	2022-06-14 → 2022-11-10	15	48	48	2	0
F074	Infoblox-Threat-Intelligence	2021-01-01 → 2026-10-15	–	–	–	–	–	–
F091	NOCACTI: Intrusion Feed	2026-09-30 → 2026-10-29	2026-09-30 → 2026-10-29	0.01	0.02	27	144	0
F092	NOCACTI: Adversary Infrastructure Feed	2025-11-23 → 2026-10-29	2026-10-01 → 2026-10-29	39	330	340	2	0
F093	Rectifyq 2024	2006-02-01 → 2023-11-13	–	–	–	–	–	–
F094	Rectifyq 2025	2022-09-24 → 2024-06-01	2024-10-24 → 2024-10-24	761	761	761	0	0
F095	Rectifyq 2026	2008-01-02 → 2026-07-29	2016-12-28 → 2026-08-01	0.05	177	6,671	3,270	0
F096	Rectifyq ICS-OT	2022-03-28 → 2024-01-17	2026-02-27 → 2026-02-28	1,431	1,431	1,431	0	0

ID	Source	first_seen range	last_seen range	Median (d)	P90 (d)	Max (d)	Zero-length	Inverted
F097	Rectifyq Malaysia	2019-05-23 → 2026-07-25	2024-08-23 → 2026-07-25	0.14	761	2,020	115	0

Table 6.2 – Range of the dates found and lifetime statistics (last_seen − first_seen, in days) for IoCs that have both dates.

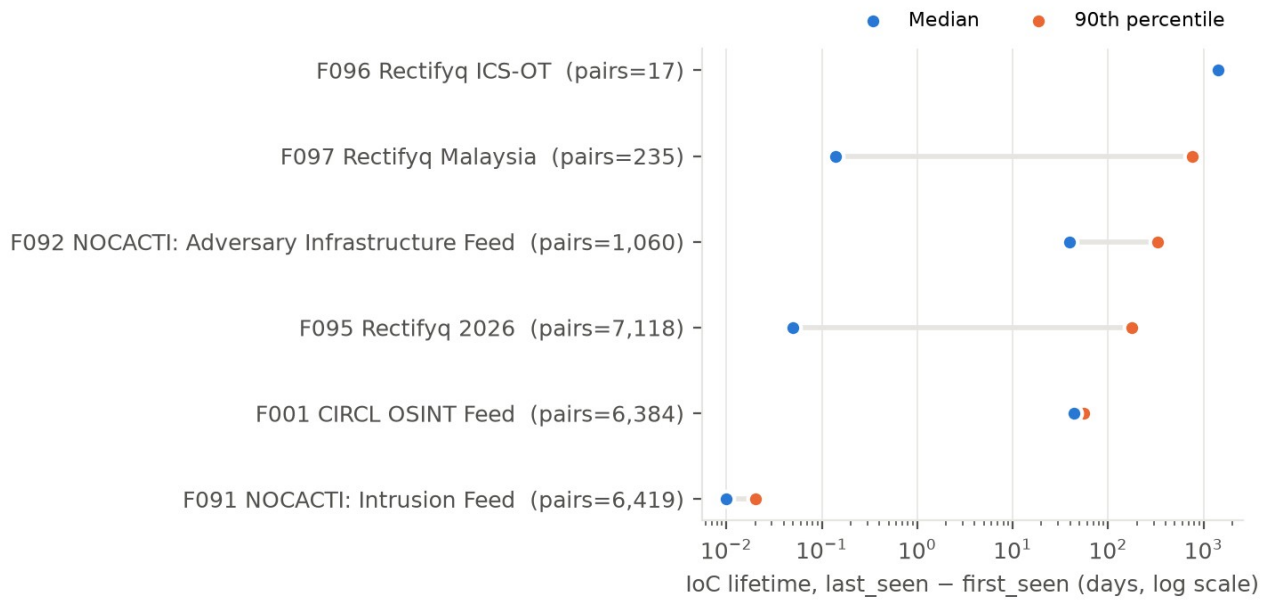


Figure 6.2 – Median and 90th percentile of IoC lifetime per source (sources with at least 10 IoCs having both dates).

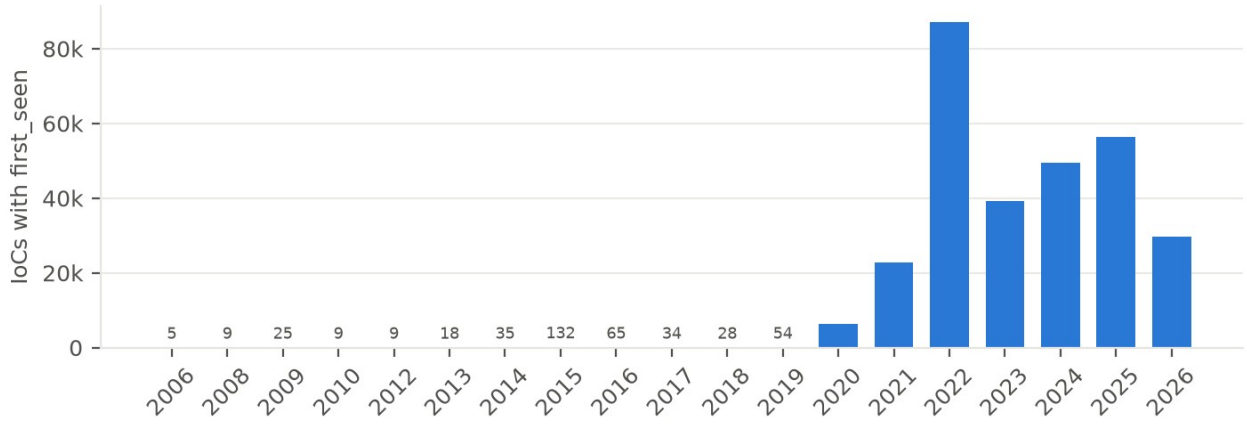


Figure 6.3 – Year of first_seen, all MISP-format sources combined.

6.3 Other information available per source

Besides attribute dates, sources differ in structure. Several put date information where Flowintel does not keep it: in object-level first_seen/last_seen (dropped by Flowintel) or in date-typed object attributes such as first-seen/last-seen (kept as plain attribute values, not as IoC dates).

ID	Source	Objects	Main IoC types	IDS flag	Object-level dates (source)	first/last-seen as object attributes (source)
F001	CIRCL OSINT Feed	23,178	domain 50.7%, hostname 7.1%, ip-dst 6.9%	83.7%	185 / 47	197 / 124
F002	The Botvrij.eu Data	762	sha256 25.2%, url 18.9%, domain 14.2%	88.1%	132 / 0	0 / 1

ID	Source	Objects	Main IoC types	IDS flag	Object-level dates (source)	first/last-seen as object attributes (source)
F074	Infoblox-Threat-Intelligence	0	domain 99.8%, ip-dst 0.1%, url 0.1%	100.0%	0 / 0	0 / 0
F083	Rösti	35,020	domain 31.3%, sha256 22.4%, url 13.2%	85.7%	0 / 0	0 / 0
F091	NOCACTI: Intrusion Feed	1,801	text 39.2%, ip-src 29.8%, counter 29.1%	28.1%	1,801 / 1,801	0 / 0
F092	NOCACTI: Adversary Infrastructure…	0	ip-dst 100.0%	100.0%	0 / 0	0 / 0
F093	Rectifyq 2024	11,751	text 16.1%, sha256 8.4%, md5 8.1%	61.9%	56 / 0	1 / 0
F094	Rectifyq 2025	18,111	domain 13.2%, hostname 10.2%, text 9.3%	81.1%	7 / 3	0 / 0
F095	Rectifyq 2026	5,848	sha256 10.9%, text 10.6%, md5 9.8%	77.6%	859 / 859	0 / 0
F096	Rectifyq ICS-OT	931	md5 12.2%, text 12.1%, sha256 11.9%	71.9%	2 / 2	0 / 0
F097	Rectifyq Malaysia	4,499	md5 12.2%, sha256 11.8%, text 11.3%	76.3%	36 / 32	0 / 0
F098	PrecisionSec OSINT	23,452	text 30.5%, datetime 21.7%, domain 11.7%	43.4%	0 / 0	14,835 / 4,792
F099	SiberKapan	0	ip-src 100.0%	100.0%	0 / 0	0 / 0
F102	TweetFeed	0	url 46.6%, domain 40.2%, ip-dst 8.0%	100.0%	0 / 0	0 / 0

Table 6.3 – Structure of each MISP-format source. Object-level first_seen/last_seen and first-seen/last-seen object attributes are counted in the raw feed.

6.4 Integrity check: source vs Flowintel

Each MISP-format source was checked by comparing the number of IoCs, first_seen and last_seen computed from the downloaded files with the values in Flowintel. After the schema workaround (section 4.2) all sources match exactly. CSV/free-text sources were checked by comparing the number of parsed IoCs with the number stored (all match).

ID	Source	IoCs in source	IoCs in Flowintel	first_seen source / Flowintel	last_seen source / Flowintel	Match
F001	CIRCL OSINT Feed	588,220	588,220	6,894 / 6,894	300,638 / 300,638	Yes
F002	The Botvrij.eu Data	28,915	28,915	329 / 329	5 / 5	Yes
F074	Infoblox-Threat-Intelligence	269,362	269,362	269,362 / 269,362	0 / 0	Yes
F083	Rösti	428,536	428,536	0 / 0	0 / 0	Yes
F091	NOCACTI: Intrusion Feed	6,419	6,419	6,419 / 6,419	6,419 / 6,419	Yes
F092	NOCACTI: Adversary Infrastructure Feed	1,060	1,060	1,060 / 1,060	1,060 / 1,060	Yes
F093	Rectifyq 2024	79,955	79,955	255 / 255	0 / 0	Yes
F094	Rectifyq 2025	224,292	224,292	2 / 2	1 / 1	Yes
F095	Rectifyq 2026	70,867	70,867	7,118 / 7,118	7,118 / 7,118	Yes
F096	Rectifyq ICS-OT	7,274	7,274	17 / 17	17 / 17	Yes
F097	Rectifyq Malaysia	43,623	43,623	236 / 236	235 / 235	Yes
F098	PrecisionSec OSINT	90,353	90,353	0 / 0	0 / 0	Yes
F099	SiberKapan	14,427	14,427	0 / 0	0 / 0	Yes
F102	TweetFeed	97,979	97,979	0 / 0	0 / 0	Yes

Table 6.4 – Counts computed directly from the downloaded feed files compared with what Flowintel stored.

- With the default schema the same comparison fails for 15 sources: 338,907 IoCs were missing, including 1,851 dated IoCs of NOCACTI Intrusion and 4,644 of Rectifyq 2026.

- Values over 255 characters: 12,872 (longest 52,679 characters) – mostly URLs, YARA/Sigma rule text and long descriptions.

7. Results – CSV and free-text sources

65 CSV/free-text sources were imported (1,569,928 IoCs, mainly IP addresses, domains and URLs). None of these IoCs has a first_seen or last_seen in Flowintel. The table also shows how many feed lines contain one or more dates, i.e. date information that the source has but that the standard MISP/Flowintel import does not use.

ID	Source	Fmt	Rows	IoCs in Flowintel	Main type	Rows with 1 date	Rows with ≥2 dates	Dates in Flowintel
F003	blockrules of rules.emergingthreats.net	csv	524	524	ip-dst	0	0	0
F004	Tor exit nodes	csv	2,063	2,063	ip-dst	0	0	0
F005	Tor ALL nodes	csv	9,838	9,838	ip-dst	0	0	0
F006	cybercrime-tracker.net - all	text	22,700	22,686	url	2	0	0
F007	Phishtank online valid phishing	csv	54,077	54,073	url	0	54,075	0
F009	diamondfox_panels	text	889	1,731	url	889	0	0
F010	pop3gropers	csv	157,414	157,414	ip-dst	0	0	0
F011	Feodo IP Blocklist	csv	6	5	ip-dst	0	5	0
F012	OpenPhish url list	text	300	300	url	0	0	0
F013	firehol_level1	text	4,657	4,657	ip-dst	0	0	0
F016	ci-badguys.txt	text	15,000	15,000	ip-dst	0	0	0
F017	alienvault reputation generic	csv	609	609	ip-dst	0	0	0
F018	blocklist.de/lists/all.txt	text	27,011	27,011	ip-dst	0	0	0
F019	VNC RFB	csv	2,416	2,416	ip-dst	2,416	0	0
F020	sshpwauth.txt	csv	10,482	10,482	ip-dst	10,482	0	0
F021	sipregistration	csv	367	367	ip-dst	367	0	0
F022	sipquery	csv	7,530	7,530	ip-dst	7,530	0	0
F023	sipinvitation	csv	101	101	ip-dst	101	0	0
F024	DNS recursion desired	csv	34,078	34,078	ip-dst	34,078	0	0
F025	DNS recursion desired IN ANY	csv	20,675	20,675	ip-dst	20,675	0	0
F026	DNS CH TXT version.bind	csv	6,365	6,365	ip-dst	6,365	0	0
F027	IP protocol 41	csv	52,163	52,163	ip-dst	0	52,163	0
F028	SMTP data	csv	422	422	ip-dst	422	0	0
F029	SMTP greet	csv	9,126	9,126	ip-dst	9,126	0	0
F030	TELNET login	csv	50,903	50,903	ip-dst	50,903	0	0
F032	VXvault - URL List	text	105	101	url	0	0	0
F033	abuse.ch SSL IPBL	csv	11,087	11,087	sha1	11,087	0	0
F034	http://cybercrime-tracker.net hashlist	text	444	444	sha1	0	0	0
F035	http://cybercrime-tracker.net gatelist	text	272	269	url	0	0	0
F036	blocklist.greensnow.co	csv	4,880	4,880	ip-dst	0	0	0
F044	ipspamlist	csv	6	5	ip-dst	0	5	0
F045	malsilo.url	csv	2	2	url	0	2	0
F046	malsilo.ipv4	csv	8	8	ip-dst	0	8	0
F047	malsilo.domain	csv	4	4	domain	0	4	0
F048	malshare.com - current all	text	12	36	md5	0	0	0
F050	IPsum (aggregation of all feeds) - level	text	144,576	144,576	ip-dst	0	0	0
F051	IPsum (aggregation of all feeds) - level	text	38,101	38,101	ip-dst	0	0	0
F052	IPsum (aggregation of all feeds) - level	text	19,707	19,707	ip-dst	0	0	0
F053	IPsum (aggregation of all feeds) - level	text	11,426	11,426	ip-dst	0	0	0
F054	IPsum (aggregation of all feeds) - level	text	5,728	5,728	ip-dst	0	0	0

ID	Source	Fmt	Rows	IoCs in Flowintel	Main type	Rows with 1 date	Rows with ≥2 dates	Dates in Flowintel
F055	IPsum (aggregation of all feeds) - level	text	2,148	2,148	ip-dst	0	0	0
F056	IPsum (aggregation of all feeds) - level	text	545	545	ip-dst	0	0	0
F057	IPsum (aggregation of all feeds) - level	text	93	93	ip-dst	0	0	0
F059	Metasploit exploits with CVE assigned	csv	3,259	3,259	vulnerability	0	0	0
F065	URL Seen in honeypots	text	2,267	2,267	url	0	0	0
F066	SSH Bruteforce IPs	csv	2,957	5,077	ip-dst	0	0	0
F067	Telnet Bruteforce IPs	csv	6,710	9,951	ip-dst	0	0	0
F069	James Brine Bruteforce IPs	csv	89,421	12,608	ip-dst	89,420	0	0
F070	List of malicious domains in Poland	text	125,533	125,533	domain	0	0	0
F075	Threatview.io - OSINT Threat Feed	text	314	314	ip-dst	0	0	0
F076	Threatview.io - C2 Hunt Feed	csv	1,178	1,378	ip-dst	0	1	0
F077	Threatview.io - IP Blocklist	text	20,909	20,907	ip-dst	0	0	0
F078	Threatview.io - Domain Blocklist	text	550,377	550,301	domain	11	0	0
F079	Threatview.io - MD5 Hash Blocklist	text	20	20	md5	0	0	0
F080	Threatview.io - URL Blocklist	text	1,632	1,632	url	0	0	0
F081	Threatview.io - Bitcoin Address Intel	text	1,392	1,388	btc	0	0	0
F082	Threatview.io - SHA File Hash Blocklist	text	31	31	sha1	0	0	0
F087	SERPRO: Reputation IP Feed	text	30,771	30,756	ip-src	0	0	0
F100	HoneyLabs active exploiters	csv	2,913	2,913	ip-dst	0	0	0
F101	HoneyLabs malware infrastructure	text	734	734	url	0	0	0
F103	Threatbase (IP feed)	text	20,000	20,000	ip-dst	0	0	0
F104	Threatbase (Domain feed)	text	20,000	20,000	domain	1	0	0
F105	Threatbase (URL feed)	text	10,000	10,000	url	0	0	0
F106	Threatbase (File hash feed)	text	20,000	20,000	sha1	0	0	0
F107	phishunt.io active phishing URLs	text	1,160	1,160	url	0	0	0

Table 7.1 – CSV/free-text sources: rows in the feed, IoCs stored in Flowintel, and rows whose raw line contains one or more date values.

- dataplane.org feeds (12 sources) have one timestamp per line (last observation); Feodo, ipspamlist and the MalSilo feeds have explicit first-seen / last-seen columns; Phishtank has submission and verification times.
- Using these dates would need a custom parser that maps the columns to first_seen/last_seen; it is not part of MISP feed handling or of Flowintel.
- James Brine (89,421 lines, 12,608 unique IPs) and the APNIC honeypot feeds contain repeated IoCs; duplicates within a source were removed.

8. Findings and recommendations

Only a minority of TI sources provide IoC lifetimes

Of the 14 usable MISP-format sources, only the two NOCACTI feeds give both first_seen and last_seen for every IoC. Infoblox gives a start date for every IoC (2021-01-01 to 2026-10-15) but no end date. Four sources (Rösti, PrecisionSec, TweetFeed, SiberKapan; 631,295 IoCs) give no attribute dates at all.

Across all MISP-format sources only 1.1% of IoCs have both dates. Statistical work on IoC lifetime is therefore possible only for a few sources, and the results are dominated by NOCACTI and Rectifyq 2026.

Dates describe very different things per source

NOCACTI Intrusion: median lifetime 0.01 days (about 15 minutes) – a single sighting. NOCACTI Adversary Infrastructure: median 39 days, P90 330 days – infrastructure activity. CIRCL OSINT: median 44 days. Rectifyq 2026: median 0.05 days but a long tail up to 18 years (6,671 days).

CIRCL OSINT has last_seen on 51% of IoCs but almost always without first_seen; these come from 288 daily 'Maltrail IOC for <date>' events where last_seen is the publication day. Counting 'end dates' without checking their origin would overstate the available lifetime information.

CSV/free-text feeds lose their dates

None of the 1.57 million CSV/free-text IoCs has a date in Flowintel, even though most lines of the dataplane.org, abuse.ch SSLBL, James Brine, Feodo, ipspamlist, MalSilo and Phishtank feeds contain dates. This follows MISP's feed model (values only), not a Flowintel defect, but it means most IP/domain reputation data enters without any temporal information.

Flowintel defects affecting TI imports

The 255-character limit on attribute values combined with per-attribute commits causes failed and half-imported cases on PostgreSQL. In this test it silently removed 9.6% of the IoCs, including dated ones, until it was worked around.

Object-level first_seen/last_seen are discarded. The bulk import path is slow (about 90 IoCs/s per process).

Recommendations

- Report to the Flowintel project: change misp__attribute.value to TEXT (with a hash or expression index), wrap create_with_event in a single transaction so that a failure does not leave partial cases, and store object-level first_seen/last_seen.
- For TI work with Flowintel, prefer sources that provide attribute first_seen/last_seen (NOCACTI, Infoblox for start dates, Rectifyq 2026, CIRCL for selected events) and document what each source's dates mean.
- If temporal analysis of reputation feeds is needed, add a custom import that maps the date columns of CSV feeds (dataplane.org, abuse.ch, Feodo, ipspamlist, MalSilo, Phishtank) to first_seen/last_seen.
- For bulk imports, batch attribute inserts in Flowintel or import through MISP and sync, and plan for several hours per million IoCs with the current code.
- Run Flowintel in production mode (./launch.sh -p, gunicorn behind nginx with TLS); do not expose the development server.
- Add requirements-dev.txt to the documented test procedure and avoid relying on an interactive shell in launch.sh.

9. Limitations

- Snapshot of 29 October 2026: feed contents change daily; CSV/free-text feeds in particular contain only current entries.
- 29 catalogue entries are not analysed: 25 could not be downloaded (gone, forbidden, timeout) and 4 returned an HTML page or empty content; some may work with registration or from another network.
- CSV/free-text parsing re-implements MISP's rules (value columns, delimiters, exclusion regex, free-text type detection); small differences in type detection compared with MISP are possible. Duplicate IoCs within a source were removed.
- Dates were analysed as stored in attribute first_seen/last_seen. Dates in event metadata, object-level fields, object attributes or comments are reported in table 6.3 but not used as IoC dates.
- The schema workaround (TEXT column, hash index, extra index, synchronous_commit=off) changes the tested system; figures for the unmodified schema are given in section 6.4.
- The development launch mode was used; production mode (gunicorn) was not tested.

Appendix A – Full source list

ID	Source	Provider	Format	Status	IoCs imported
F000	ELLIO: IP Feed (Community version)	ellio.tech	freetext	Downloaded, not usable	–
F001	CIRCL OSINT Feed	CIRCL	misp	Imported	588,220
F002	The Botvrij.eu Data	Botvrij.eu	misp	Imported	28,915
F003	blockrules of rules.emergingthreats.net	rules.emergingthreats.net	csv	Imported	524
F004	Tor exit nodes	TOR Node List from dan.me.uk -	csv	Imported	2,063
F005	Tor ALL nodes	TOR Node List from dan.me.uk -	csv	Imported	9,838
F006	cybercrime-tracker.net - all	cybercrime-tracker.net	freetext	Imported	22,686
F007	Phishtank online valid phishing	Phishtank	csv	Imported	54,073
F008	ip-block-list - snort.org	https://snort.org	freetext	Downloaded, not usable	–
F009	diamondfox_panels	pan-unit42	freetext	Imported	1,731
F010	pop3gropers	home.nuug.no	csv	Imported	157,414
F011	Feodo IP Blocklist	abuse.ch	csv	Imported	5
F012	OpenPhish url list	openphish.com	freetext	Imported	300
F013	firehol_level1	iplists.firehol.org	freetext	Imported	4,657
F014	IPs from High-Confidence DGA-Based C&Cs Actively Resolving -	osint.bambenekconsulting.com	csv	Unavailable	–
F015	Domains from High-Confidence DGA-based C&C Domains Actively	osint.bambenekconsulting.com	csv	Unavailable	–
F016	ci-badguys.txt	cinsscore.com	freetext	Imported	15,000
F017	alienvault reputation generic	.alienvault.com	csv	Imported	609
F018	blocklist.de/lists/all.txt	blocklist.de	freetext	Imported	27,011
F019	VNC RFB	dataplane.org	csv	Imported	2,416
F020	sshpwauth.txt	dataplane.org	csv	Imported	10,482
F021	sipregistration	dataplane.org	csv	Imported	367
F022	sipquery	dataplane.org	csv	Imported	7,530
F023	sipinvitation	dataplane.org	csv	Imported	101
F024	DNS recursion desired	dataplane.org	csv	Imported	34,078
F025	DNS recursion desired IN ANY	dataplane.org	csv	Imported	20,675
F026	DNS CH TXT version.bind	dataplane.org	csv	Imported	6,365
F027	IP protocol 41	dataplane.org	csv	Imported	52,163
F028	SMTP data	dataplane.org	csv	Imported	422
F029	SMTP greet	dataplane.org	csv	Imported	9,126
F030	TELNET login	dataplane.org	csv	Imported	50,903
F031	All current domains belonging to known malicious DGAs	osint.bambenekconsulting.com	csv	Unavailable	–
F032	VXvault - URL List	VXvault	freetext	Imported	101
F033	abuse.ch SSL IPBL	abuse.ch	csv	Imported	11,087
F034	http://cybercrime-tracker.net hashlist	http://cybercrime-tracker.net	freetext	Imported	444
F035	http://cybercrime-tracker.net gatelist	http://cybercrime-tracker.net	freetext	Imported	269
F036	blocklist.greensnow.co	greensnow.co	csv	Imported	4,880
F037	This list contains all domains - A list for administrators t	ZeroDot1 - CoinBlockerLists	freetext	Unavailable	–
F038	This list contains all optional domains - An additional list	ZeroDot1 - CoinBlockerLists	freetext	Unavailable	–
F039	This list contains all browser mining domains - A list to pr	ZeroDot1 - CoinBlockerLists	freetext	Unavailable	–
F040	URLHaus Malware URLs	abuse.ch	csv	Unavailable	–
F041	CyberCure - IP Feed	www.cybercure.ai	csv	Unavailable	–
F042	CyberCure - Blocked URL Feed	www.cybercure.ai	csv	Unavailable	–
F043	CyberCure - Hash Feed	www.cybercure.ai	csv	Unavailable	–
F044	ipspamlist	ipspamlist	csv	Imported	5

ID	Source	Provider	Format	Status	IoCs imported
F045	malsilo.url	MalSilo	csv	Imported	2
F046	malsilo.ipv4	MalSilo	csv	Imported	8
F047	malsilo.domain	MalSilo	csv	Imported	4
F048	malshare.com - current all	malshare.com	freetext	Imported	36
F049	Panels Tracker	Benkow.cc	csv	Unavailable	–
F050	IPsum (aggregation of all feeds) - level 1 - lot of false po	IPsum	freetext	Imported	144,576
F051	IPsum (aggregation of all feeds) - level 2 - medium false po	IPsum	freetext	Imported	38,101
F052	IPsum (aggregation of all feeds) - level 3 - low false posit	IPsum	freetext	Imported	19,707
F053	IPsum (aggregation of all feeds) - level 4 - very low false	IPsum	freetext	Imported	11,426
F054	IPsum (aggregation of all feeds) - level 5 - ultra false pos	IPsum	freetext	Imported	5,728
F055	IPsum (aggregation of all feeds) - level 6 - no false positi	IPsum	freetext	Imported	2,148
F056	IPsum (aggregation of all feeds) - level 7 - no false positi	IPsum	freetext	Imported	545
F057	IPsum (aggregation of all feeds) - level 8 - no false positi	IPsum	freetext	Imported	93
F058	DigitalSide Threat-Intel OSINT Feed	osint.digitalside.it	misp	Unavailable	–
F059	Metasploit exploits with CVE assigned	eCrimeLabs	csv	Imported	3,259
F060	Malware Bazaar	abuse.ch	csv	Unavailable	–
F061	PhishScore	PhishStats	csv	Unavailable	–
F062	Threatfox	abuse.ch	misp	Unavailable	–
F063	MalwareBazaar	abuse.ch	misp	Unavailable	–
F064	URLhaus	abuse.ch	misp	Unavailable	–
F065	URL Seen in honeypots	APNIC Community Honeynet Proje	freetext	Imported	2,267
F066	SSH Bruteforce IPs	APNIC Community Honeynet Proje	csv	Imported	5,077
F067	Telnet Bruteforce IPs	APNIC Community Honeynet Proje	csv	Imported	9,951
F068	threatfox indicators of compromise	abuse.ch	csv	Unavailable	–
F069	James Brine Bruteforce IPs	jamesbrine.com.au	csv	Imported	12,608
F070	List of malicious domains in Poland	CERT-PL	freetext	Imported	125,533
F071	List of malicious hashes	Banco do Brasil S.A	csv	Unavailable	–
F072	Shreshta: Newly Registered domain names(NRD) - 1 week (Commu	shreshtait.com	freetext	Downloaded, not usable	–
F073	Shreshta: Newly Registered domain names (NRD) - 1 month (Com	shreshtait.com	freetext	Downloaded, not usable	–
F074	Infoblox-Threat-Intelligence	infoblox.com	misp	Imported	269,362
F075	Threatview.io - OSINT Threat Feed	threatview.io	freetext	Imported	314
F076	Threatview.io - C2 Hunt Feed	threatview.io	csv	Imported	1,378
F077	Threatview.io - IP Blocklist	threatview.io	freetext	Imported	20,907
F078	Threatview.io - Domain Blocklist	threatview.io	freetext	Imported	550,301
F079	Threatview.io - MD5 Hash Blocklist	threatview.io	freetext	Imported	20
F080	Threatview.io - URL Blocklist	threatview.io	freetext	Imported	1,632
F081	Threatview.io - Bitcoin Address Intel	threatview.io	freetext	Imported	1,388
F082	Threatview.io - SHA File Hash Blocklist	threatview.io	freetext	Imported	31
F083	Rösti - Repackaged Öpen Source Threat Intelligence	bin.re	misp	Imported	428,536
F084	Phishing.Database - New domains of today	Phishing.Database	csv	Unavailable	–
F085	Phishing.Database - New IPs of today	Phishing.Database	csv	Unavailable	–
F086	Phishing.Database - New URLs of today	Phishing.Database	csv	Unavailable	–
F087	SERPRO: Reputation IP Feed	SERPRO	freetext	Imported	30,756
F088	hideNseek LAB - Threat Intelligence JSON	hideNseek LAB	freetext	Unavailable	–
F089	hideNseek LAB - CSV Threat Feed	hideNseek LAB	csv	Unavailable	–
F090	hideNseek LAB - IP Blocklist	hideNseek LAB	freetext	Unavailable	–

ID	Source	Provider	Format	Status	IoCs imported
F091	NOCACTI: Intrusion Feed	NOCACTI	misp	Imported	6,419
F092	NOCACTI: Adversary Infrastructure Feed	NOCACTI	misp	Imported	1,060
F093	Rectifyq - Curated Enriched MISP Threat Information for 2024	rectifyq.com	misp	Imported	79,955
F094	Rectifyq - Curated Enriched MISP Threat Information for 2025	rectifyq.com	misp	Imported	224,292
F095	Rectifyq - Curated Enriched MISP Threat Information for 2026	rectifyq.com	misp	Imported	70,867
F096	Rectifyq - Curated Enriched MISP Threat Information for ICS-	rectifyq.com	misp	Imported	7,274
F097	Rectifyq - Curated Enriched MISP Threat Information relevant	rectifyq.com	misp	Imported	43,623
F098	PrecisionSec OSINT Threat Intelligence	precisionsec.com	misp	Imported	90,353
F099	SiberKapan	siberkapan.org	misp	Imported	14,427
F100	HoneyLabs active exploiters	HoneyLabs	csv	Imported	2,913
F101	HoneyLabs malware infrastructure	HoneyLabs	freetext	Imported	734
F102	TweetFeed	tweetfeed.live	misp	Imported	97,979
F103	Threatbase (IP feed)	threatbase.qzz.io	freetext	Imported	20,000
F104	Threatbase (Domain feed)	threatbase.qzz.io	freetext	Imported	20,000
F105	Threatbase (URL feed)	threatbase.qzz.io	freetext	Imported	10,000
F106	Threatbase (File hash feed)	threatbase.qzz.io	freetext	Imported	20,000
F107	phishunt.io active phishing URLs	phishunt.io	freetext	Imported	1,160

Appendix B – Commands and scripts

Download all feeds (VM)

```
python3 -I tools/fi_fetch_feeds.py defaults.json feeds
```

Native statistics from the raw feeds

```
python3 -I tools/fi_raw_stats.py feeds        # -> raw_stats.json
```

Import into Flowintel (one process per source group and app instance)

```
python3 -I tools/fi_import_feeds.py feeds --key <API key> --url http://127.0.0.1:7006 --only <ids> --log
import_X.jsonl
```

Schema workaround (test only)

```
DROP INDEX ix_misp__attribute_value;
ALTER TABLE misp__attribute ALTER COLUMN value TYPE text;
CREATE INDEX ix_misp__attribute_value ON misp__attribute USING hash (value);
CREATE INDEX ix_misp__attribute_case_misp_object_id ON misp__attribute (case_misp_object_id);
ALTER SYSTEM SET synchronous_commit = off;
```

Repair / verification of incomplete cases

```
flowintel/env/bin/python -I tools/fi_repair_partial.py feeds --key <API key> --url http://127.0.0.1:7006 --only <ids>
```

Statistics from the Flowintel database (read-only)

```
flowintel/env/bin/python tools/fi_stats.py      # -> flowintel_stats.json
```

Charts and report

```
venv/bin/python make_summary.py data out ; python3 make_notes.py out/summary.json facts.json
out/notes.json ; node gen_report.js out report.docx
```